

SIH REPORT

Low Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration

Problem Statement ID: SIH 26064

Theme: Robotics and Drones

Category: Hardware

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Abstract

Write a concise summary of the proposed system here. Describe the problem, proposed solution, key methodology, expected outcome, and significance of the project.

1. Introduction

1.1 Background

Introduce the broader problem and explain why it is important.

1.2 Problem Statement

Clearly describe the problem that the proposed project aims to solve.

1.3 Motivation

Explain why this project is necessary and what motivated the proposed solution.

1.4 Objectives

The main objectives of the proposed system are:

1. Objective 1.
2. Objective 2.
3. Objective 3.
4. Objective 4.

2. Existing Solutions and Literature Review

2.1 Existing Technologies

Discuss currently available technologies and approaches.

2.2 Limitations of Existing Solutions

Explain the limitations of current systems.

2.3 Research Gap

Describe what is missing from existing approaches and how your project addresses that gap.

3. Proposed Solution

3.1 Overview

Describe your proposed system at a high level.

3.2 Mission Concept of Operations

Explain how the system is deployed and operated throughout a mission.

3.3 Advantages of the Proposed Approach

tell what are the benefits in our proposed model and workflow

4. System Design and Selection

There are three major subsystem in ANVESHAK.

- Mechanical Subsystem
- Electrical Subsystem
- Software for processing

A structured walkthrough about the architecture of these subsystems are given in this chapter.

4.1 Mechanical Design

Describe the mechanical structure of the system.

subsystem list

then material selection

then

4.1.1 Material Selection

Discuss the materials considered and the final selection (with alternative, reason of choice, cost and advantage for all (both selected and alternative))

4.1.2 Pressure Housing

Describe the pressure resistant enclosure.

4.1.3 Sensor Housing

Describe the sensor mounting arrangement.

4.2 Electronic Design

Describe the electronic architecture.

4.2.1 Sensors

List and describe the sensors used.(with alternative, reason of choice, cost and advantage for all (both selected and alternative))

4.2.2 Microcontroller

Describe the selected microcontroller and its role.

4.2.3 Power System

Describe the battery and power management system.

4.3 Software Design

Describe the software architecture and algorithms. write in a contextual form.(start from, receiving data from the ANVESHAK and then preprocessing all the way down to conclusion, what kind of data is expected from ANVESHAK and how you have chosen a specific model for a specific role)

try to focus more on design part, because a proper detailed workflow will be added on section 5.4

make a flowchart if possible, landscape format, and contact rabinath before updating the figure.

5. Sensor and Detection Methodology

5.1 Detection Principle

Explain the physical principle used for detecting metal deposits.

5.2 Sensor Configuration

Describe the geometry and arrangement of the sensors. This might look similar to section 4.1.3 but here you will give your proper detailed and scientifically backed argument for the choice of sensor housing unlike in section 4.1.3 where it was focused on just the mounting part with less descriptive reasons of choice.

5.3 Signal Processing

Explain how the raw sensor signal is processed.

5.4 Detection Algorithm

Describe the algorithm used for identifying metallic targets.

This might look similar as section 4.3 but here you will focus more on the data processing and detection part than the design part which is already mentioned in the 4.3 section in details.

6. Deployment Strategy

6.1 Deployment Mechanism

Explain how the payload will be deployed underwater.

use flowchart or figures.

since its a proposed strategy, you can add your own method instead of conventional methods, like using a small vessel(unmanned boat) to deliver the ANVESHAK to a particular target location on the ocean surface.

6.2 Depth Requirement

Discuss the target operating depth.

here add a table comparing the capability of ANVESHAK vs the requirement for a particular(use an example location and depth) location.

6.3 Recovery Mechanism

Describe how the system will return to the surface, all the way upto recovering the data.

7. Design Calculations

IMPORTANT BUT WE W

7.1 Pressure Calculation

Present the pressure calculations required for the target depth. FOR NOW, GIVE THE CALCULATIONS CONSIDERING THE ANVESHAK AS A UNIFORM SPHERICAL ENCLOSURE.

7.2 Structural Analysis

ONCE THE COMPLETE MODEL IS BUILT, ONLY THEN IT CAN BE UPDATED.

FIRST SOFTWARE BASED ANALYSIS

LATER THE TEST REPORTS THROUGH THE PROTOTYPE.

Discuss the structural requirements of the enclosure.

7.3 Power Budget

Table 7.1: Estimated power consumption of major components.

Component	Power (W)	Quantity
Sensor-1	—	—
Sensor-2	—	—
Microcontroller	—	—
Communication System	—	—
Other Electronics	—	—

7.4 Mass Budget

Table 7.2: Estimated mass of major components.

Component	Power (W)	Quantity
Sensor-1	—	—
Sensor-2	—	—
Microcontroller	—	—
Other Electronic	—	—
mounting system	—	—
enclosure	—	—

8. Cost Analysis

Table 8.1: Estimated project cost. for one model

Component	Quantity	Estimated Cost (INR)	Buffer
Sensor	1	—	—
Microcontroller	1	—	—
Battery	1	—	—
Pressure Housing	1	—	—
Electronics	—	—	—
Miscellaneous	—	—	—
Total		—	—

Table 8.2: Estimated cost. for prototyping and testing

Component	Quantity	Estimated Cost (INR)	Buffer
Sensor	1	—	—
Microcontroller	1	—	—
Battery	1	—	—
Pressure Housing	1	—	—
Electronics	—	—	—
Miscellaneous	—	—	—
Total		—	—

9. Expected Results

9.1 Expected Performance

Describe the expected performance of the system.

9.2 Expected Detection Range

Discuss the expected detection range and sensitivity.

9.3 Advantages

similar to section 3.3 but here, instead of overview, write it in context of the section 9.1

- Low-cost implementation.
- Deployable underwater.
- Modular design.

10. Challenges and Risk Analysis

THIS CHAPTER SHOULD BE FILLED BY ALL THE TEAMMMATES BECAUSE REQUIRES EVERY FIELD.

10.1 Technical Challenges

Discuss major technical challenges.

10.2 Environmental Challenges

Discuss underwater pressure, corrosion, temperature, salinity, etc.

10.3 Risk Mitigation

Describe how each major risk will be addressed.

11. Future Scope

Discuss possible improvements and future extensions. THIS CAN BE UPDATED LATER WHEN WE HAVE DONE RESONABLE WORK ON THE MODEL ALREADY AND HAVE DONE SOME PROTOTYPING TOO.

Possible future developments include:

- Autonomous navigation.
- Real-time communication.
- AI-assisted detection.
- Improved sensor sensitivity.
- Integration with an underwater remotely operated vehicle.

12. Conclusion

Summarize the proposed system, its significance, expected performance, and potential impact.

LATER :)

Bibliography

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- [2] Author Name, “Title of Research Paper,” *Journal Name*, vol. X, no. X, pp. XX–XX, Year.
- [3] Organization, *Technical Documentation Title*, Year.